

- 1 The photograph shows a brine shrimp, *Artemia salina*.



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Magnification $\times 2.0$

Brine shrimps are invertebrate animals that live in salt lakes. They feed on algae and reproduce by producing eggs.

When the conditions are favourable, the eggs hatch into brine shrimps.

A student observed that eggs hatched faster when calcium ions were added to the saltwater.

(a) Describe an experiment to investigate the effect of calcium ion concentration on the time taken for brine shrimps to hatch.

(5)

(b) (i) State **two abiotic** variables, other than the independent variable, that could affect this experiment.

(2)

First variable.....

Second variable.....

(b) (ii) Choose **one** of the variables you have identified in (b)(i).

Explain how this variable could be controlled.

Describe what effect it could have on the results if it is not controlled.

(2)

Variable

How this variable is controlled.

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Effect it could have on the results if it is not controlled.

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(c) Calcium ions can bind to some enzymes.

Explain how the presence of calcium ions can affect the time taken for brine shrimp eggs to hatch.

(3)

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(Total for Question 1 = 12 marks)
